

Proof H is NOT Sequentially Consistent: Proof by contradiction.

Assume H is linearizable. As H is a complete history (without pending invocations), this means there exists a history S such that:

- (a) S is equivalent to H ;
- (b) S is a legal sequential history.

For S to be a legal history as per (b) we must have: ... (1)

From (a) and definition of H we have: ... (2)

From (1) and (2) we have: ...

That is ... and thus $\rightarrow_S$ contains a cycle. This leads to a contradiction as S is a sequential history and thus $\rightarrow_S$ must be a strict (irreflexive) total order.